

Lab 03 - Service Station Control System Emulation



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Lattes

ORCID

SciProfiles

Scopus

ResearcherID

substationworm

LFreitasGutierrez

Tasks

Warning:

When using specialized search engines or Google dorks to identify Internet-exposed OT devices, **do not interact with or attempt to access any real systems**. The tasks in this document are strictly educational, observational, and non-intrusive, and must fully comply with ethical and legal standards.

- **1** Verify the IP address of the `otlab-student` workstation.
- **2** Determine the subnet range of the network where the `otlab-student` workstation is deployed.
- **3** Discover the IP address, MAC address, and vendor information of other active hosts within the network.
 - *Hint: One of the devices is an automatic tank gauge (ATG) controller.*
- **4** Identify open ports and available services on the OT-ICS host over both TCP and UDP protocols.
- **5** Retrieve additional system information using vendor-specific `nmap` scripts.
- **6** Identify the total number of publicly accessible OT-ICS devices using the same open port as `gas_station` through a specialized search engine such as [Shodan](#) or [FOFA](#).
- **7** Determine the total number of publicly exposed OT-ICS devices that implement the same ATG-related functionality as the `gas_station` host by querying a specialized search engine such as [Shodan](#) or [FOFA](#).

Tools

- The following tools are available on the `otlab-student` workstation for completing OT-Lab 3: `ifconfig`, `masscan`, `netdiscover`, and `nmap`.

Nomenclature

- ATG: Automatic tank gauge.
- ICS: Industrial control system.
- IP: Internet protocol.
- MAC: Media access control.
- OT: Operational technology.
- TCP: Transmission control protocol.
- UDP: User datagram protocol.